
Multi-Agent Owner-Assisted Scoring Mechanisms

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1 Introduction

A fundamental challenge in many domains is to elicit high-quality information from owners to help with value evaluation when directly verifying the acquired information is not feasible, either because the ground truth is unavailable or because it is too costly to obtain. Recently, this problem has drawn more and more attention from conference committees. For instance, ICML 2023¹ has decided to use the Isotonic Mechanism [2], according to Prof. Weijie Su from Wharton School, University of Pennsylvania.

The owner-assisted scoring problem is a newly emerging algorithmic game theory problem with little prior work. Su [1] initially defines this problem and gives a feasible truthful mechanism for the single-agent setting. Furthermore, Su [2] gives more discussion about noise assumptions and makes the proofs more intelligible. However, these two works only consider the case of a single agent. They suppose that all objects belong to one owner and the final goal of the owner is to maximize the sum of the value of all her objects after taking a convex utility function. Such a setting excludes the possibility of eliciting information from different owners about each respective object.

An example is not far to seek that we may consider the GDP reports of all provinces. While the central government observes only a noisy estimation, they need to elicit more information from the local government about the truthful GDP numbers and use them to improve the estimations. On the other hand, each provincial government get funds according to the refined estimated GDP. Hence, the GDP estimated problem has become a game because each provincial government wants to maximize its reward funds by manipulating the reports to the central government.

This paper will be the first study on the multi-agent owner-assisted scoring problem. The main question we explore is to design a mechanism incorporating observed noisy values and owners' reports to incentivize truthful reports and a better estimation simultaneously. We will first give a truthful mechanism for two owners and then extend it to more general cases.

2 Models

In our multi-agent owner-assisted problem, there are n agents (owners) $A_1, A_2, \dots, A_n$. Each agent A_i owns an item with true value R_i . For convenience, we denote $(R_i)_{i \in [n]}$ by $\mathbf{R}$. The principal can observe a noisy or even biased estimation $Y_i \sim \mathcal{Y}_i$ of R_i . Also for convenience, we denote $(Y_i)_{i \in [n]}$ by $\mathbf{Y}$. For sufficient information that Y_i contains, we have the following assumption.

Assumption 2.1 (Consistency of $\mathbf{Y}$). For any $i, j \in [n]$, $\mathbb{E}[Y_i - Y_j] > 0$ if $R_i > R_j$.

According to theorem 1 in [2], the most precise information the principal can elicit from agents truthfully is the ranking of their items in descending order of the ground-truth values $R_1, \dots, R_n$. Therefore, we let the principal ask each agent for pair-wise comparison between her item and others. From all the truthful pair-wise comparisons, one can get the truthful ranking of all the items and

¹<https://icml.cc/Conferences/2023/Dates>

hence improve the estimations. We denote the improved estimations of $R_1, \dots, R_n$ to be $\hat{Y}_1, \dots, \hat{Y}_n$. For convenience, we furthermore denote $(\hat{Y}_i)_{i \in [n]}$ by $\hat{\mathbf{Y}}$. It is worth noting that the mechanism of computing $\hat{Y}_1, \dots, \hat{Y}_n$ needs to be made public for fairness and transparency.

Formally, a mechanism $\mathcal{M}$ for our problem is defined by a function M that takes $Y_1, \dots, Y_n$ and some pair-wise comparisons as inputs. M outputs $\hat{Y}_1, \dots, \hat{Y}_n$ such that when all pair-wise comparisons are truthful, $\hat{Y}_1, \dots, \hat{Y}_n$ is more close to $R_1, R_2, \dots, R_n$ than the original $Y_1, \dots, Y_n$.

Each agent A_i has a private utility function U_i for the estimated value of her own item. When $\hat{Y}_i$ is given by the principal, the reward of A_i is $U_i(\hat{Y}_i)$. For simplicity, we temporarily set all U_i 's as monotone increasing linear functions.

3 A Truthful Scoring Mechanism for Two Agents

In this section, we consider the case for two agents A_1 and A_2 . As we introduced in section 2, the principal asks both of them a same question whether $R_1 > R_2$ or $R_1 < R_2$. We suppose that the two agents answers $\oplus_1, \oplus_2 \in \{>, < \}$. Each of them represents the value of one agent's item compared to the value of the another one. Now we design our mechanism $\mathcal{M}_2$ for A_1 and A_2 . We let $\hat{Y}_1$ and $\hat{Y}_2$ as below:

$$\hat{Y}_1 = \begin{cases} Y_1 & \text{if } \oplus_1 \neq \oplus_2 \text{ and } Y_1 \oplus_1 Y_2 \\ \frac{Y_1 + Y_2}{2} & \text{if } \oplus_1 \neq \oplus_2 \text{ but } \neg(Y_1 \oplus_1 Y_2) , \\ Y_1 - \frac{1}{2}(Y_2 - Y_1)\mathbb{I}[Y_1 < Y_2] & \text{if } \oplus_1 = \oplus_2 \end{cases} \quad (1)$$

$$\hat{Y}_2 = \begin{cases} Y_2 & \text{if } \oplus_1 \neq \oplus_2 \text{ and } Y_1 \oplus_1 Y_2 \\ \frac{Y_1 + Y_2}{2} & \text{if } \oplus_1 \neq \oplus_2 \text{ but } \neg(Y_1 \oplus_1 Y_2) . \\ Y_2 - \frac{1}{2}(Y_1 - Y_2)\mathbb{I}[Y_2 < Y_1] & \text{if } \oplus_1 = \oplus_2 \end{cases} \quad (2)$$

With this mechanism, it is easy to verify that when A_1 and A_2 are truthful, $\|\hat{\mathbf{Y}} - \mathbf{R}\|_l \leq \|\mathbf{Y} - \mathbf{R}\|_l, l \in \mathbb{N}^+$. Now we prove that if agent A_1 and A_2 are both rational and know that others are rational, they will report truthfully. Formally, we have the following theorem 3.1. Due to the space limitation, the proof of theorem 3.1 has been put into appendix B.

Theorem 3.1 (Truthfulness of $\mathcal{M}_2$). *In the owner-assisted scoring model introduced in section 2, if both agents are rational and know that others are rational, mechanism $\mathcal{M}_2$ guarantees that both agents report truthfully.*

Although theorem 3.1 already gives us a persuasive reason to use $\mathcal{M}_2$ in our scoring problem, we can ulteriorly prove the same result for a more general family of utility functions U_i 's. However, the range of feasible U_i 's depends on the minimal difference $\kappa_0^{\mathbf{Y}}$ between Y_i 's. Formally, we have the following definition for $\kappa_0^{\mathbf{Y}}$.

Definition 3.2 (Minimal Difference of Original Estimator $\mathbf{Y}$). $\kappa_0^{\mathbf{Y}}$ is defined by

$$\kappa_0^{\mathbf{Y}} = \min_{i,j \in [n], R_i > R_j} \frac{\int_0^{+\infty} \Pr\{Y_i - Y_j > c\}dc}{\int_0^{+\infty} \Pr\{Y_j - Y_i > c\}dc}. \quad (3)$$

Remark: We give a justification of $\kappa_0^{\mathbf{Y}}$ here. For simplicity, we denote $\frac{\int_0^{+\infty} \Pr\{Y_i - Y_j > c\}dc}{\int_0^{+\infty} \Pr\{Y_j - Y_i > c\}dc}$ as $\kappa_{i,j}^{\mathbf{Y}}$.

Therefore we can alternatively write $\kappa_{i,j}^{\mathbf{Y}}$ as $\frac{\mathbb{E}[Y_i - Y_j | Y_i > Y_j]}{-\mathbb{E}[Y_i - Y_j] + \mathbb{E}[Y_i - Y_j | Y_i > Y_j]}$. According to assumption 2.1, $\kappa_{i,j}^{\mathbf{Y}} > 1$, which implies that $\kappa_0^{\mathbf{Y}} > 1$ according to eq. (3). In appendix A, we have demonstrated that given $Y_i = R_i + \epsilon_i, \epsilon_i \sim \mathcal{N}(0, \sigma)$ for all $i \in [n]$, the relationship between $\min_{i \neq j} |R_i - R_j|$ and $\max \kappa_0^{\mathbf{Y}}$.

Then we define a family of functions.

Definition 3.3 (Generalized Bilipschitz Functions). A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is a generalized $\sqrt{\kappa}$ -bilipschitz function if and only if there exist two positive constants κ_1, κ_2 such that for any $x, y \in \mathbb{R}$,

$$\kappa_1 |x - y| \leq |f(x) - f(y)| \leq \kappa_2 |x - y|,$$

and $\frac{\kappa_2}{\kappa_1} \leq \kappa$. When $\kappa_1 = \frac{1}{\sqrt{\kappa}}$ and $\kappa_2 = \sqrt{\kappa}$, f is exactly a $\sqrt{\kappa}$ -bilipschitz function.

With these preliminaries, we can propose an even more persuasive theorem that works for monotone-increasing generalized bilipschitz functions. Due to the space limitation, the proof of theorem 3.4 has been put into appendix C.

Theorem 3.4 (Truthfulness of $\mathcal{M}_2$ for Extended Utility Functions). *In the model we introduced in section 2, if $\mathbf{Y}$ further satisfies $\kappa_0^{\mathbf{Y}} \geq \kappa_0$ in definition 3.2 and both agents A_1, A_2 are rational and know that others are rational, mechanism $\mathcal{M}_2$ guarantees that both agents report truthfully when U_i 's are monotone increasing generalized $\sqrt{\kappa_0}$ -bilipschitz functions.*

Remark: In appendix C, we prove that with $\kappa_0^{\mathbf{Y}} \geq \kappa_0$, mechanism $\mathcal{M}_2$ is feasible for $\sqrt{\kappa_0}$ -bilipschitz utility functions. Conversely, if $\kappa_0^{\mathbf{Y}}$ is smaller than κ_0 , we can still prove that our new estimator $\hat{\mathbf{Y}}$ is still not much worse than the original estimator $\mathbf{Y}$. Intuitively, if $\kappa_0^{\mathbf{Y}}$ is very insignificant, $\mathbb{E}[Y_1]$ and $\mathbb{E}[Y_2]$ will be very close in the two agents setting, thus an incorrect comparison between R_1, R_2 will not push $\hat{\mathbf{Y}}$ away from $\mathbf{R}$ too much. Formally, we have

Lemma 3.5. *In the model we introduced in section 2, we let $Y_i = R_i + \epsilon_i$ where ϵ_i is sampled from a zero-mean distribution. If $\kappa_0^{\mathbf{Y}}$ is not bounded as definition 3.2 defines, the estimator $\hat{\mathbf{Y}}$ generated by $\mathcal{M}_2$ satisfies that $\mathbb{E}[\|\hat{\mathbf{Y}} - \mathbf{R}\|_1 - \|\mathbf{Y} - \mathbf{R}\|_1] < \frac{1}{2} \max\{\delta_{1,2}(\kappa_0), \delta_{2,1}(\kappa_0)\}$ when agents are using monotone increasing generalized $\sqrt{\kappa_0}$ -bilipschitz functions. Here $\delta_{i,j}(\kappa_0)$ is defined by the maximal δ such that $\delta + (\kappa_0 - 1) \int_{c=0}^{\delta} \Pr\{\epsilon_i - \epsilon_j > c\}dc \leq \frac{\kappa_0 - 1}{2} \int_{c=0}^{+\infty} \Pr\{|\epsilon_i - \epsilon_j| > c\}dc$.*

Proof. When $\kappa_0^{\mathbf{Y}} \geq \kappa_0$, it is easy to verify that $\|\hat{\mathbf{Y}} - \mathbf{R}\|_1 \leq \|\mathbf{Y} - \mathbf{R}\|_1$. We only consider the case that $\kappa_0^{\mathbf{Y}} < \kappa_0$. According to eq. (1) and eq. (2), we can deduce that

$$\mathbb{E}[\|\hat{\mathbf{Y}} - \mathbf{R}\|_1 - \|\mathbf{Y} - \mathbf{R}\|_1] \leq \mathbb{E}[\|\hat{\mathbf{Y}} - \mathbf{Y}\|_1] \leq \mathbb{E}\left[\left|\frac{Y_1 - Y_2}{2}\right|\right].$$

Without loss of generality, we suppose that $R_1 \geq R_2$. If $\mathbb{E}[|Y_1 - Y_2|] > \max\{\delta_{1,2}(\kappa_0), \delta_{2,1}(\kappa_0)\}$, we can deduce that

$$\begin{aligned} \kappa_0^{\mathbf{Y}} &= \frac{\int_0^{+\infty} \Pr\{Y_1 - Y_2 > c\}dc}{\int_0^{+\infty} \Pr\{Y_2 - Y_1 > c\}dc} \\ &= \frac{\int_0^{+\infty} \Pr\{\epsilon_1 - \epsilon_2 > c - (R_1 - R_2)\}dc}{\int_0^{+\infty} \Pr\{\epsilon_2 - \epsilon_1 > c - (R_2 - R_1)\}dc} \\ &= \frac{\frac{1}{2} \int_{c=0}^{+\infty} \Pr\{|\epsilon_1 - \epsilon_2| > c\}dc + (R_1 - R_2) - \int_{c=0}^{R_1 - R_2} \Pr\{\epsilon_2 - \epsilon_1 > c\}}{\frac{1}{2} \int_{c=0}^{+\infty} \Pr\{|\epsilon_1 - \epsilon_2| > c\}dc - \int_{c=0}^{R_1 - R_2} \Pr\{\epsilon_2 - \epsilon_1 > c\}} \\ &\geq \kappa_0, \end{aligned}$$

which is a contradiction. Hence, our claim is proven. $\square$

Now we have demonstrated that $\hat{\mathbf{Y}}$ will not get much worse than $\mathbf{Y}$. In appendix A, we will also show the relation between $\delta_{i,j}(\kappa_0)$ and κ_0 when the noises ϵ_i 's are sampled from a normal distribution.

4 A Truthful Scoring Mechanism for Multiple Agents

In this section, we extend our mechanism $\mathcal{M}_2$ for two agents to multiple agents and prove its truthfulness. Intuitively, if we see the improved estimation $\hat{\mathbf{Y}}$ as rewards to the agents, we can perform $\mathcal{M}_2$ on each pair of agents and take an average of all of these rewards. Then each agent should be willing to perform well in each sub-game and report truthfully. Suppose the principal asks pairs $P_1, \dots, P_m$ about their pair-wise comparisons. For an agent $A_i \in P_j$, denote the report of her by $\oplus_{i,j}$ and her estimated value under $\mathcal{M}_2$ applying on pair P_j by $M_2(Y_i, Y_k, \oplus_{i,j}, \oplus_{j,k})$ where $A_k \in P_j$. With these notations, we define mechanism $\mathcal{M}_n$ for agents $A_1, \dots, A_n$ as below:

$$\hat{Y}_i = \frac{1}{n-1} \left(\left(n-1 - \sum_{j=1}^m \mathbb{I}[A_i \in P_j] \right) Y_i + \sum_{j,k: P_j=(A_i, A_k)} M_2(Y_i, Y_k, \oplus_{i,j}, \oplus_{j,k}) \right). \quad (4)$$

103 In appendix D, we show that $\hat{\mathbf{Y}}$ defined by eq. (4) also satisfies $\|\hat{\mathbf{Y}} - \mathbf{R}\|_2 \leq \|\mathbf{Y} - \mathbf{R}\|_2$. Then
 104 similar to theorem 3.1, we can also get a theorem to show the truthfulness of $\mathcal{M}_n$.

105 **Theorem 4.1** (Truthfulness of $\mathcal{M}_n$). *In the owner-assisted scoring model introduced in section 2,*
 106 *if all n agents are rational and know that others are rational, mechanism $\mathcal{M}_n$ guarantees that all*
 107 *agents report truthfully in all pair-wise comparisons.*

108 *Proof.* According to eq. (4) we can deduce that in order to maximize her own utility, each agent A_i
 109 should maximize each $M_2(Y_i, Y_k, \oplus_{i,j}, \oplus_{j,k})$. Then in a pair-wise comparison $P_j = (A_i, A_k)$, if
 110 $Y_i > Y_k$, according to eq. (4), we have

$$\begin{aligned} \mathbb{E}[M_2(Y_i, Y_k, \oplus_{i,j} = >, \oplus_{j,k} = <)] &> \mathbb{E}[M_2(Y_i, Y_k, \oplus_{i,j} = <, \oplus_{j,k} = <)], \\ \mathbb{E}[M_2(Y_i, Y_k, \oplus_{i,j} = >, \oplus_{j,k} = >)] &> \mathbb{E}[M_2(Y_i, Y_k, \oplus_{i,j} = <, \oplus_{j,k} = >)], \\ \mathbb{E}[M_2(Y_k, Y_i, \oplus_{j,k} = >, \oplus_{i,j} = <)] &> \mathbb{E}[M_2(Y_k, Y_i, \oplus_{j,k} = <, \oplus_{i,j} = <)]. \end{aligned}$$

111 Hence, if A_i and A_k are both rational and know that everyone is rational, they will select $\oplus_{i,j} = >$
 112 $, \oplus_{j,k} = >$ to maximize both of their utilities. Therefore, we can conclude that all agents should be
 113 truthful in all pair-wise comparisons. $\square$

114 Similar to theorem 3.4, we can also prove a more general theorem of truthfulness for $\mathcal{M}_n$. Due to the
 115 space limitation, the proof of theorem 4.2 has been put into appendix E.

116 **Theorem 4.2** (Truthfulness of $\mathcal{M}_2$ for Extended Utility Functions). *In the model we introduced in*
 117 *section 2, if $\mathbf{Y}$ further satisfies $\kappa_0^{\mathbf{Y}} = \kappa_0$ in definition 3.2 and all agents $A_1, A_2, \dots, A_n$ are rational*
 118 *and know that others are rational, mechanism $\mathcal{M}_n$ guarantees that all agents report truthfully when*
 119 *U_i 's are monotone increasing generalized $\sqrt{\kappa_0}$ -bilipschitz functions.*

120 5 An Extension for Multiple Agents with Multiple Items Belong to Each of 121 Them

122 In our previous settings, each agent only has one item. However, there are some circumstances
 123 such that each agent has multiple items. It is easy to verify that our mechanism $\mathcal{M}_n$ still achieve
 124 truthfulness but we may want a strongly truthful mechanism. A feasible way to do this is to slightly
 125 modify our mechanism $\mathcal{M}_2$ for two agents to $\mathcal{M}'_2$.

$$\hat{Y}_1 = \begin{cases} Y_1 & \text{if } \oplus_1 \neq \oplus_2 \text{ and } Y_1 \oplus_1 Y_2 \\ \frac{(1+\epsilon)Y_1 + (1-\epsilon)Y_2}{2} & \text{if } > = \oplus_1 \neq \oplus_2 \text{ but } \neg(Y_1 \oplus_1 Y_2) \\ \frac{(1-\epsilon)Y_1 + (1+\epsilon)Y_2}{2} & \text{if } < = \oplus_1 \neq \oplus_2 \text{ but } \neg(Y_1 \oplus_1 Y_2) \\ Y_1 - \frac{1}{2}(Y_2 - Y_1)\mathbb{I}[Y_1 < Y_2] & \text{if } \oplus_1 = \oplus_2 \end{cases}, \quad (5)$$

$$\hat{Y}_2 = \begin{cases} Y_2 & \text{if } \oplus_1 \neq \oplus_2 \text{ and } Y_1 \oplus_1 Y_2 \\ \frac{(1+\epsilon)Y_1 + (1-\epsilon)Y_2}{2} & \text{if } > = \oplus_1 \neq \oplus_2 \text{ but } \neg(Y_1 \oplus_1 Y_2) \\ \frac{(1-\epsilon)Y_1 + (1+\epsilon)Y_2}{2} & \text{if } < = \oplus_1 \neq \oplus_2 \text{ but } \neg(Y_1 \oplus_1 Y_2) \\ Y_2 - \frac{1}{2}(Y_1 - Y_2)\mathbb{I}[Y_2 < Y_1] & \text{if } \oplus_1 = \oplus_2 \end{cases}. \quad (6)$$

126 Here, ϵ is a small positive constant less than $\frac{1}{2}$. It can be proven that $\mathcal{M}'_2$ is strongly truthful for linear
 127 utility functions U_i 's. Formally, we have the following theorem proven in appendix F.

128 **Theorem 5.1** (Truthfulness of $\mathcal{M}'_n$). *In the owner-assisted scoring model introduced in section 2,*
 129 *if each agent has multiple items and all n agents are rational and know that others are rational,*
 130 *mechanism $\mathcal{M}'_n$ guarantees that all agents report truthfully in all pair-wise comparisons.*

131 However, it remains an open problem that whether $\hat{\mathbf{Y}}$ produced by this mechanism is a good
 132 estimation of $\mathbf{R}$.

133 6 A Comparison Between $\mathcal{M}_n$ and Isotonic Mechanism

134 In [2], Isotonic regression is used to get a new estimator $\hat{\mathbf{Y}}'$ of $\mathbf{R}$. In this section, we compare $\hat{\mathbf{Y}}$
 135 generated by our mechanism $\mathcal{M}_n$ with $\hat{\mathbf{Y}}'$ generated by Isotonic regression.

136 We randomly generate $\mathbf{R} \sim \mathcal{U}^{50}[0, 1]$ and each $Y_i = R_i + \epsilon_i$ where $\epsilon_i \sim \mathcal{N}[0, \sigma]$ for $i \in [50]$. Then
 137 we use Isotonic regression and $\mathcal{M}_n$ on all the pair-wise comparisons and then compute the two-norm
 138 loss $\|\mathbf{Y} - \mathbf{R}\|$, $\|\hat{\mathbf{Y}} - \mathbf{R}\|$ and $\|\hat{\mathbf{Y}}' - \mathbf{R}\|$. We run this simulation for each σ for 100 times and
 139 compute the average two-norm losses with their ranges. The result is shown in fig. 1.

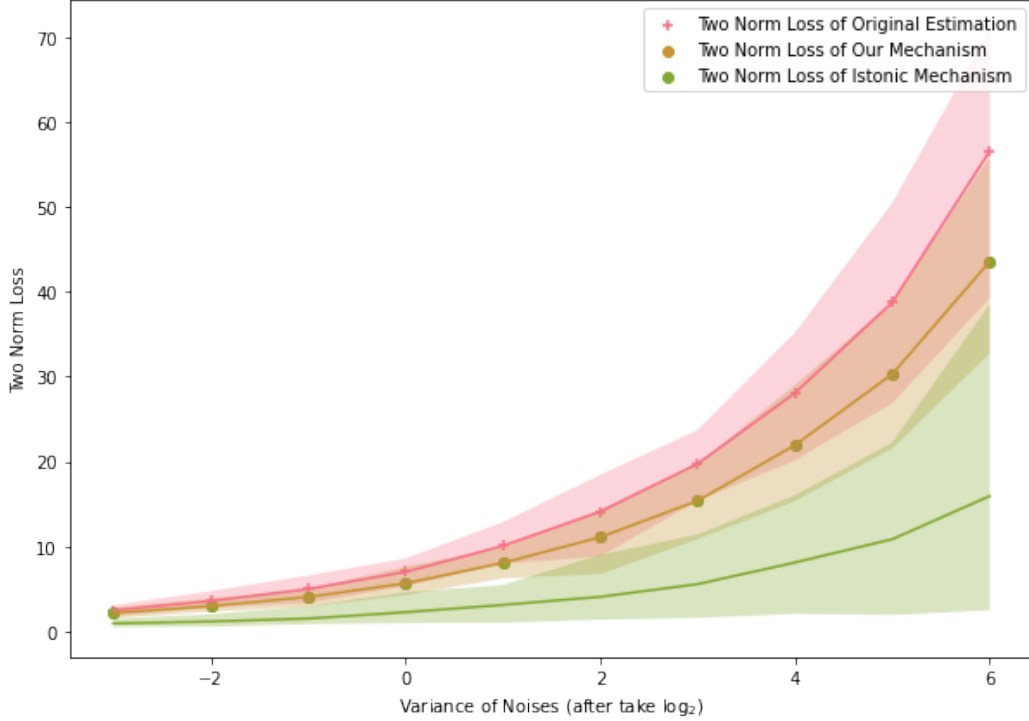


Figure 1: A comparison of performance between Isotonic regression and our mechanism.

140 We can see that our mechanism indeed gets better estimation than the original $\mathbf{Y}$ but is still outper-
 141 formed by Isotonic regression. Whether a truthful mechanism with better performance than $\mathcal{M}_n$ is
 142 left as a future work.

143 7 Future Plan

144 The framework of our paper has already been built up. However, there are still some details and
 145 extensions that need to be made in the next few weeks. Firstly, we need to write a justification
 146 for definition 3.2 and show that a relatively large $\kappa_0^{\mathbf{Y}}$ usually exists. This part will be included in
 147 appendix A. Secondly, there are still some lemmas that are not proven formally. Although they are
 148 not very difficult, we still need some time to complete them in appendix D and appendix F. Thirdly,
 149 as we can see in section 6, the performance of our mechanism is worse than the Isotonic mechanism
 150 for the single-agent setting. We need a more detailed comparison and try to justify this.

151 Furthermore, we plan to conduct a social experiment among some students. We will perform the
 152 game and mechanism for multiple rounds and observe the behavioral dynamics of different people.

153 References

- 154 [1] W. Su. You are the best reviewer of your own papers: An owner-assisted scoring mechanism.
 155 *Advances in Neural Information Processing Systems*, 34:27929–27939, 2021.
 156 [2] W. J. Su. A truthful owner-assisted scoring mechanism. *arXiv preprint arXiv:2206.08149*, 2022.

157 Appendix

158 A Justifications of Definition 3.2

159 This part is left as a future work.

160 B Proof of Theorem 3.1

161 *Proof.* Without loss of generality, we suppose that $R_1 > R_2$. The expected rewards of both agents
 162 are represented by $S_{\oplus_1 \oplus_2}^1$ and $S_{\oplus_2 \oplus_1}^2$ if they report $\oplus_1, \oplus_2$ respectively. Because of rationality, we
 163 only need to prove that $S_{><}^1 > S_{<<}^1$, $S_{>>}^1 > S_{<>}^1$ and $S_{><}^2 > S_{<<}^2$, then they will decide to
 164 truthfully report $\oplus_1 = >, \oplus_2 = <$. Actually, by the linearity of U_i , we have

$$\begin{aligned} S_{><}^1 - S_{<<}^1 &= \mathbb{E}[U_1(Y_1) | Y_1 > Y_2] \Pr\{Y_1 > Y_2\} \\ &\quad + \mathbb{E}\left[U_1\left(\frac{Y_1 + Y_2}{2}\right) \middle| Y_1 \leq Y_2\right] \Pr\{Y_1 \leq Y_2\} \\ &\quad - \mathbb{E}[U_1(Y_1)] + \mathbb{E}\left[U_1\left(\frac{Y_2 - Y_1}{2}\right) \middle| Y_1 \leq Y_2\right] \Pr\{Y_1 \leq Y_2\} \\ &= 2\mathbb{E}\left[U_1\left(\frac{Y_2 - Y_1}{2}\right) \middle| Y_1 \leq Y_2\right] > 0, \end{aligned}$$

165 and

$$\begin{aligned} S_{>>}^1 - S_{<>}^1 &= -\mathbb{E}[U_1(Y_1) | Y_1 \leq Y_2] \Pr\{Y_1 \leq Y_2\} \\ &\quad - \mathbb{E}\left[U_1\left(\frac{Y_1 + Y_2}{2}\right) \middle| Y_1 > Y_2\right] \Pr\{Y_1 > Y_2\} \\ &\quad + \mathbb{E}[U_1(Y_1)] - \mathbb{E}\left[U_1\left(\frac{Y_2 - Y_1}{2}\right) \middle| Y_1 \leq Y_2\right] \Pr\{Y_1 \leq Y_2\} \\ &= \mathbb{E}\left[U_1\left(\frac{Y_1 - Y_2}{2}\right)\right] > 0, \end{aligned}$$

166 and

$$\begin{aligned} S_{><}^2 - S_{<<}^2 &= \mathbb{E}[U_2(Y_2) | Y_1 > Y_2] \Pr\{Y_1 > Y_2\} \\ &\quad + \mathbb{E}\left[U_2\left(\frac{Y_1 + Y_2}{2}\right) \middle| Y_1 \leq Y_2\right] \Pr\{Y_1 \leq Y_2\} \\ &\quad - \mathbb{E}[U_2(Y_2)] + \mathbb{E}\left[U_2\left(\frac{Y_1 - Y_2}{2}\right) \middle| Y_1 > Y_2\right] \Pr\{Y_1 > Y_2\} \\ &= \mathbb{E}\left[U_2\left(\frac{Y_1 - Y_2}{2}\right)\right] > 0. \end{aligned}$$

167 Until now, theorem 3.1 has been proven. □

168 C Proof of Theorem 3.4

169 *Proof.* Without loss of generality, we only consider the case of $R_1 > R_2$. Suppose the corresponding
 170 κ_1, κ_2 in definition 3.3 for U_i are $\kappa_{1,i}, \kappa_{2,i}$. Same as the proof of theorem 3.1, we only need to prove
 171 that $S_{><}^1 > S_{<<}^1$, $S_{>>}^1 > S_{<>}^1$ and $S_{><}^2 > S_{<<}^2$, then both of them will decide to truthfully report
 172 $\oplus_1 = >, \oplus_2 = <$. The following proofs are different from those of theorem 3.1 because U_i 's are no

173 longer linear. Actually, we have

$$\begin{aligned}
S_{><}^1 - S_{<<}^1 &= \mathbb{E}[U_1(Y_1)|Y_1 > Y_2] \Pr\{Y_1 > Y_2\} \\
&\quad + \mathbb{E}\left[U_1\left(\frac{Y_1 + Y_2}{2}\right)\middle|Y_1 \leq Y_2\right] \Pr\{Y_1 \leq Y_2\} \\
&\quad - \mathbb{E}[U_1(Y_1)|Y_1 > Y_2] \Pr\{Y_1 > Y_2\} - \mathbb{E}\left[U_1\left(\frac{3Y_1 - Y_2}{2}\right)\middle|Y_1 \leq Y_2\right] \Pr\{Y_1 \leq Y_2\} \\
&= \mathbb{E}\left[U_1\left(\frac{Y_1 + Y_2}{2}\right) - U_1\left(\frac{3Y_1 - Y_2}{2}\right)\middle|Y_1 \leq Y_2\right] \Pr\{Y_1 \leq Y_2\} > 0,
\end{aligned}$$

174 and

$$\begin{aligned}
S_{>>}^1 - S_{<>}^1 &= -\mathbb{E}[U_1(Y_1)|Y_1 \leq Y_2] \Pr\{Y_1 \leq Y_2\} \\
&\quad - \mathbb{E}\left[U_1\left(\frac{Y_1 + Y_2}{2}\right)\middle|Y_1 > Y_2\right] \Pr\{Y_1 > Y_2\} \\
&\quad + \mathbb{E}[U_1(Y_1)|Y_1 > Y_2] \Pr\{Y_1 > Y_2\} + \mathbb{E}\left[U_1\left(\frac{3Y_1 - Y_2}{2}\right)\middle|Y_1 \leq Y_2\right] \Pr\{Y_1 \leq Y_2\} \\
&= \mathbb{E}\left[U_1(Y_1) - U_1\left(\frac{Y_1 + Y_2}{2}\right)\middle|Y_1 > Y_2\right] \Pr\{Y_1 > Y_2\} \\
&\quad - \mathbb{E}\left[U_1(Y_1) - U_1\left(\frac{3Y_1 - Y_2}{2}\right)\middle|Y_1 \leq Y_2\right] \Pr\{Y_1 \leq Y_2\} \\
&\geq \kappa_{1,1} \mathbb{E}\left[\frac{Y_1 - Y_2}{2}\middle|Y_1 > Y_2\right] \Pr\{Y_1 > Y_2\} - \kappa_{2,1} \mathbb{E}\left[\frac{Y_2 - Y_1}{2}\middle|Y_1 \leq Y_2\right] \Pr\{Y_1 \leq Y_2\} \\
&= \frac{\kappa_{1,1}}{2} \int_0^{+\infty} \Pr\{Y_1 - Y_2 > c\} dc - \frac{\kappa_{2,1}}{2} \int_0^{+\infty} \Pr\{Y_2 - Y_1 > c\} dc \\
&= \frac{1}{2} \left(\kappa_{1,1} \int_0^{+\infty} \Pr\{Y_1 - Y_2 > c\} dc - \kappa_{2,1} \int_0^{+\infty} \Pr\{Y_2 - Y_1 > c\} dc \right) > 0,
\end{aligned}$$

175 and

$$\begin{aligned}
S_{><}^2 - S_{<<}^2 &= \mathbb{E}[U_2(Y_2)|Y_1 > Y_2] \Pr\{Y_1 > Y_2\} \\
&\quad + \mathbb{E}\left[U_2\left(\frac{Y_1 + Y_2}{2}\right)\middle|Y_1 \leq Y_2\right] \Pr\{Y_1 \leq Y_2\} \\
&\quad - \mathbb{E}[U_2(Y_2)|Y_1 \leq Y_2] \Pr\{Y_1 \leq Y_2\} - \mathbb{E}\left[U_2\left(\frac{3Y_2 - Y_1}{2}\right)\middle|Y_1 > Y_2\right] \Pr\{Y_1 > Y_2\} \\
&= \mathbb{E}\left[U_2(Y_2) - U_2\left(\frac{3Y_2 - Y_1}{2}\right)\middle|Y_1 > Y_2\right] \Pr\{Y_1 > Y_2\} \\
&\quad - \mathbb{E}\left[U_2(Y_2) - U_2\left(\frac{Y_1 + Y_2}{2}\right)\middle|Y_1 \leq Y_2\right] \Pr\{Y_1 \leq Y_2\} \\
&\geq \kappa_{1,2} \mathbb{E}\left[\frac{Y_1 - Y_2}{2}\middle|Y_1 > Y_2\right] \Pr\{Y_1 > Y_2\} - \kappa_{2,2} \mathbb{E}\left[\frac{Y_2 - Y_1}{2}\middle|Y_1 \leq Y_2\right] \Pr\{Y_1 \leq Y_2\} \\
&> 0.
\end{aligned}$$

176 Until now, theorem 3.4 has been proven. □

177 **D The Proof That $\hat{Y}$ is a Better Estimator Than Y**

178 This part is left as a future work.

179 **E Proof of Theorem 4.2**

180 *Proof.* We consider a pair of agents $P_{j_0} = (A_{i_0}, A_{j_0})$ and fix all the reports of other pairs. After
 181 fixing all of them, we denote

$$\frac{1}{n-1} \left(\left(n-1 - \sum_{j=1}^m \mathbb{I}[A_i \in P_j] \right) Y_i + \sum_{j,k: P_j=(A_{i_0}, A_k), j \neq j_0} M_2(Y_{i_0}, Y_k, \oplus_{i_0,j}, \oplus_{j,k}) \right)$$

182 by $C_{i_0,j_0}(\mathbf{Y})$. We only consider the impact of P_j 's reports on the expected utilities of agent A_{i_0}
 183 and A_{k_0} . In order to prove truthfulness, we only need to prove that if $Y_{i_0} > Y_{k_0}$, the following
 184 inequalities hold:

$$\begin{aligned} \mathbb{E} \left[U_{i_0} \left(C_{i_0,j_0}(\mathbf{Y}) + \frac{1}{n-1} M_2(Y_{i_0}, Y_{k_0}, >, <) \right) \right] &\geq \mathbb{E} \left[U_{i_0} \left(C_{i_0,j_0}(\mathbf{Y}) + \frac{1}{n-1} M_2(Y_{i_0}, Y_{k_0}, <, <) \right) \right], \\ \mathbb{E} \left[U_{i_0} \left(C_{i_0,j_0}(\mathbf{Y}) + \frac{1}{n-1} M_2(Y_{i_0}, Y_{k_0}, >, >) \right) \right] &\geq \mathbb{E} \left[U_{i_0} \left(C_{i_0,j_0}(\mathbf{Y}) + \frac{1}{n-1} M_2(Y_{i_0}, Y_{k_0}, <, >) \right) \right], \\ \mathbb{E} \left[U_{k_0} \left(C_{k_0,j_0}(\mathbf{Y}) + \frac{1}{n-1} M_2(Y_{k_0}, Y_{i_0}, >, <) \right) \right] &\geq \mathbb{E} \left[U_{k_0} \left(C_{k_0,j_0}(\mathbf{Y}) + \frac{1}{n-1} M_2(Y_{k_0}, Y_{i_0}, <, <) \right) \right]. \end{aligned}$$

185 Suppose the corresponding κ_1, κ_2 in definition 3.3 for U_i are $\kappa_{1,i}, \kappa_{2,i}$. Firstly, we have

$$\begin{aligned} &\mathbb{E} \left[U_{i_0} \left(C_{i_0,j_0}(\mathbf{Y}) + \frac{1}{n-1} M_2(Y_{i_0}, Y_{k_0}, \oplus_{i_0,j_0}, \oplus_{i_0,k_0}) \right) \right] \\ &= \mathbb{E} \left[U_{i_0} \left(C_{i_0,j_0}(\mathbf{Y}) + \frac{1}{n-1} M_2(Y_{i_0}, Y_{k_0}, \oplus_{i_0,j_0}, \oplus_{i_0,k_0}) \right) \middle| Y_{i_0} \leq Y_{k_0} \right] \Pr\{Y_{i_0} \leq Y_{k_0}\} \\ &\quad + \mathbb{E} \left[U_{i_0} \left(C_{i_0,j_0}(\mathbf{Y}) + \frac{1}{n-1} M_2(Y_{i_0}, Y_{k_0}, \oplus_{i_0,j_0}, \oplus_{i_0,k_0}) \right) \middle| Y_{i_0} > Y_{k_0} \right] \Pr\{Y_{i_0} > Y_{k_0}\}. \end{aligned}$$

186 Therefore, we can deduce that

$$\begin{aligned} &\mathbb{E} \left[U_{i_0} \left(C_{i_0,j_0}(\mathbf{Y}) + \frac{1}{n-1} M_2(Y_{i_0}, Y_{k_0}, >, <) \right) \right] - \mathbb{E} \left[U_{i_0} \left(C_{i_0,j_0}(\mathbf{Y}) + \frac{1}{n-1} M_2(Y_{i_0}, Y_{k_0}, <, <) \right) \right] \\ &= \mathbb{E} \left[U_{i_0} \left(C_{i_0,j_0}(\mathbf{Y}) + \frac{Y_{i_0} + Y_{k_0}}{2(n-1)} \right) - U_{i_0} \left(C_{i_0,j_0}(\mathbf{Y}) + \frac{3Y_{i_0} - Y_{k_0}}{2(n-1)} \right) \middle| Y_{i_0} \leq Y_{k_0} \right] \Pr\{Y_{i_0} \leq Y_{k_0}\} \\ &\quad + \mathbb{E} \left[U_{i_0} \left(C_{i_0,j_0}(\mathbf{Y}) + \frac{Y_{i_0}}{n-1} \right) - U_{i_0} \left(C_{i_0,j_0}(\mathbf{Y}) + \frac{Y_{i_0}}{n-1} \right) \middle| Y_{i_0} > Y_{k_0} \right] \Pr\{Y_{i_0} > Y_{k_0}\} \\ &> 0, \end{aligned}$$

187 and

$$\begin{aligned} &\mathbb{E} \left[U_{i_0} \left(C_{i_0,j_0}(\mathbf{Y}) + \frac{1}{n-1} M_2(Y_{i_0}, Y_{k_0}, >, >) \right) \right] - \mathbb{E} \left[U_{i_0} \left(C_{i_0,j_0}(\mathbf{Y}) + \frac{1}{n-1} M_2(Y_{i_0}, Y_{k_0}, <, >) \right) \right] \\ &= \mathbb{E} \left[U_{i_0} \left(C_{i_0,j_0}(\mathbf{Y}) + \frac{3Y_{i_0} - Y_{k_0}}{2(n-1)} \right) - U_{i_0} \left(C_{i_0,j_0}(\mathbf{Y}) + \frac{Y_{i_0}}{n-1} \right) \middle| Y_{i_0} \leq Y_{k_0} \right] \Pr\{Y_{i_0} \leq Y_{k_0}\} \\ &\quad + \mathbb{E} \left[U_{i_0} \left(C_{i_0,j_0}(\mathbf{Y}) + \frac{Y_{i_0}}{n-1} \right) - U_{i_0} \left(C_{i_0,j_0}(\mathbf{Y}) + \frac{Y_{i_0} + Y_{k_0}}{2(n-1)} \right) \middle| Y_{i_0} > Y_{k_0} \right] \Pr\{Y_{i_0} > Y_{k_0}\} \\ &\geq \mathbb{E} \left[\kappa_{2,i} \frac{Y_{i_0} - Y_{k_0}}{2(n-1)} \middle| Y_{i_0} \leq Y_{k_0} \right] \Pr\{Y_{i_0} \leq Y_{k_0}\} + \mathbb{E} \left[\kappa_{1,i} \frac{Y_{i_0} - Y_{k_0}}{2(n-1)} \middle| Y_{i_0} > Y_{k_0} \right] \Pr\{Y_{i_0} > Y_{k_0}\} \\ &> 0, \end{aligned}$$

188 where the last inequality is similarly proven as the proof of theorem 3.4, and

$$\begin{aligned}
& \mathbb{E} \left[U_{k_0} \left(C_{k_0, j_0}(\mathbf{Y}) + \frac{1}{n-1} M_2(Y_{k_0}, Y_{i_0}, >, <) \right) \right] - \mathbb{E} \left[U_{k_0} \left(C_{k_0, j_0}(\mathbf{Y}) + \frac{1}{n-1} M_2(Y_{k_0}, Y_{i_0}, <, <) \right) \right] \\
&= \mathbb{E} \left[U_{k_0} \left(C_{k_0, j_0}(\mathbf{Y}) + \frac{Y_{i_0} + Y_{k_0}}{2(n-1)} \right) - U_{k_0} \left(C_{k_0, j_0}(\mathbf{Y}) + \frac{Y_{k_0}}{n-1} \right) \middle| Y_{i_0} \leq Y_{k_0} \right] \Pr\{Y_{i_0} \leq Y_{k_0}\} \\
&+ \mathbb{E} \left[U_{k_0} \left(C_{k_0, j_0}(\mathbf{Y}) + \frac{Y_{k_0}}{n-1} \right) - U_{k_0} \left(C_{k_0, j_0}(\mathbf{Y}) + \frac{3Y_{k_0} - Y_{i_0}}{2(n-1)} \right) \middle| Y_{i_0} > Y_{k_0} \right] \Pr\{Y_{i_0} > Y_{k_0}\} \\
&\geq \mathbb{E} \left[\kappa_{1,i} \frac{Y_{i_0} - Y_{k_0}}{2(n-1)} \middle| Y_{i_0} > Y_{k_0} \right] \Pr\{Y_{i_0} > Y_{k_0}\} + \mathbb{E} \left[\kappa_{2,i} \frac{Y_{i_0} - Y_{k_0}}{2(n-1)} \middle| Y_{i_0} \leq Y_{k_0} \right] \Pr\{Y_{i_0} \leq Y_{k_0}\} \\
&> 0,
\end{aligned}$$

189 where the last inequality is also similarly proven as the proof of theorem 3.4. Therefore, we can
190 conclude that all agents should be truthful in all pair-wise comparisons. $\square$

191 **F Proof of Truthfulness of $\mathcal{M}'_n$**

192 This part is left as a future work.